Assignment 2

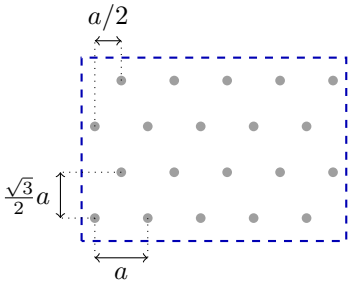
Multiscale Modelling and Computation (APL410/AML7200)

Due date: 1 November 2026
Total Marks: 40

Summary. Write a molecular-statics code in two dimensions: pair energy and forces, periodic boundary conditions, a Verlet neighbour list, and energy minimization by steepest descent and conjugate gradient. Use it to compute the formation energies and relaxation of a vacancy (Q1) and an interstitial (Q2) in a triangular Lennard–Jones crystal, to follow an edge dislocation in a square lattice under quasi-static shear until it glides out of the specimen (Q3), and to compare the cost of the N^2 and Verlet-list neighbour searches and of the two minimizers (Q4). Background material is in `supplementary_note.pdf`.

Q1: Vacancy formation and relaxation (10 marks)

Consider a two-dimensional triangular lattice with lattice spacing $a = 1.116153$. A periodic cell consists of n horizontal rows of n atoms: atoms in a row are a apart, rows are $\sqrt{3}a/2$ apart, and every odd row is shifted by $a/2$, so that $L_x = na$ and $L_y = n\sqrt{3}a/2$.



periodic cell (dashed)
sketch: 4 rows of 5 atoms

Two atoms at distance r interact through the Lennard–Jones pair potential, in reduced units ($\epsilon = \sigma = 1$),

$$\phi_{\mathrm{LJ}}(r) = 4\left(r^{-12} - r^{-6}\right). \tag{1}$$

Truncate it at $r_{\mathrm{cut}} = 2.5$ with the cutoff correction discussed in class, which makes both the potential and its derivative vanish at r_{cut}:

$$\phi(r) = \begin{cases} \phi_{\mathrm{LJ}}(r) - \phi_{\mathrm{LJ}}(r_{\mathrm{cut}}) - (r - r_{\mathrm{cut}})\,\phi'_{\mathrm{LJ}}(r_{\mathrm{cut}}), & r \le r_{\mathrm{cut}}, \\ 0, & r > r_{\mathrm{cut}}. \end{cases} \tag{2}$$

The total potential energy is $\mathcal{V} = \sum_{\alpha<\beta} \phi(r^{\alpha\beta})$, and the force on atom α is $\mathbf{f}^\alpha = -\partial\mathcal{V}/\partial\mathbf{r}^\alpha$, which includes the derivative of the linear correction term in (2). Write the energy and force evaluation and the nearest-image convention for periodic boundary conditions (PBC) yourself. The value of a above minimizes the energy of the perfect lattice for (2).

Use `perfect_20x20.csv` and `vacancy_20x20.csv`, with PBC in x and y and the cell dimensions held fixed.

(a) Calculate the total energy and the maximum atomic force of the perfect and the vacancy configurations. Verify that the force on every atom of the perfect crystal is approximately zero. **[2 marks]**

(b) Relax the vacancy configuration with conjugate gradient and line minimization, until $\max_\alpha \|\mathbf{f}^\alpha\|$, the largest force magnitude over all atoms, is below a small tolerance of your choice (for example 10^{-5}); state the tolerance. If a new CG direction is not downhill ($\mathbf{f} \cdot \mathbf{d} \le 0$), reset $\mathbf{d} = \mathbf{f}$. Plot the total energy and the maximum atomic force against CG iteration. **[3 marks]**

(c) Compute the vacancy formation energy before and after relaxation,

$$E_f^v = E_{N-1}^v - \frac{N-1}{N} E_N, \tag{3}$$

where E_N is the energy of the perfect crystal of N atoms and E_{N-1}^v that of the vacancy cell. Plot the displacement vectors of the atoms near the vacancy (magnify them and state the factor), and the radial displacement against distance from the vacant site. Explain the value of the unrelaxed E_f^v and the size of the relaxation. **[3 marks]**

(d) Repeat the relaxed calculation with `vacancy_10x10.csv` and `vacancy_16x16.csv` and compare the three values of (3). **[2 marks]**

Q2: Interstitial formation (8 marks)

Use the lattice and the potential (2) of Q1, with PBC, and `interstitial_20x20.csv`. The extra atom (last line of the file) starts at the centre of a triangle of nearest neighbours.

(a) Calculate the initial total energy and maximum atomic force, then relax with conjugate gradient as in Q1(b). Plot the maximum atomic force against CG iteration. **[2 marks]**

(b) Compute the interstitial formation energy before and after relaxation,

$$E_f^i = E_{N+1}^i - \frac{N+1}{N} E_N, \tag{4}$$

with E_N from Q1(a). **[2 marks]**

(c) Plot the atoms near the interstitial before and after relaxation, with first-neighbour bonds (lines joining atoms closer than $1.37a$). Does the inserted atom stay at the centre of its triangle? Plot the horizontal displacement u_x of the atoms in the close-packed row through its final position against their position along the row, and describe the arrangement. **[2 marks]**

(d) Compare (4) with (3), and the largest relaxation displacements of the two defects. Explain the difference using the local atomic arrangement and the interatomic forces. **[2 marks]**

Q3: Edge dislocation under quasi-static shear (14 marks)

This question uses a square lattice, with atoms at (ia, ja) and $a = 1.103090$, and the pair potential

$$\phi_{\mathrm{sq}}(r) = \phi_{\mathrm{LJ}}(r) - \epsilon_2 \exp\left[-\frac{(r-r_2)^2}{2s^2}\right], \qquad \epsilon_2 = 1, \quad r_2 = 1.56, \quad s = 0.1. \tag{5}$$

truncated as in (2), with ϕ_{LJ} replaced by ϕ_{sq}.

Use `edge_dislocation_initial.csv` (1022 atoms, **no PBC**). The atoms are labelled `bottom` (two lowest rows), `top` (two highest rows) and `mobile`; the left and right surfaces are free. The upper half contains one extra half-column of atoms (ids in `parameters.json`), so the block contains an edge dislocation with Burgers vector $\mathbf{b} = (a, 0)$ on the slip plane $y = 0$. The coordinates come from the elastic displacement field of the dislocation and are *not* relaxed. The `top` and `bottom` atoms form the constrained set C_d of the lectures: they keep their prescribed positions, and their forces are set to zero inside the minimizer.

To locate the dislocation core, use the coordination number: the number of neighbours of an atom closer than $1.2a$ (4 in the perfect square lattice). Ignore atoms within $2.5a$ of the free surfaces and the rows next to the `top` and `bottom` groups.

(a) Hold `top` and `bottom` fixed and relax the `mobile` atoms with conjugate gradient as in Q1(b). Plot the configuration before and after relaxation with first-neighbour bonds (atoms closer than $1.2a$), the extra half-column and the atoms whose coordination is not 4 highlighted. Where is the core after relaxation? **[4 marks]**

(b) Starting from the relaxed configuration, displace the `top` group horizontally in increments $\Delta u = 0.02$ up to $u = 0.5$: $x^\alpha = X^\alpha + u$ for every `top` atom, with X^α its relaxed position. Keep `bottom` fixed. After each increment, relax the `mobile` atoms starting from the previous relaxed configuration. Tabulate the number of CG iterations and the final maximum atomic force for each increment. Use $H = \langle y \rangle_{\text{top}} - \langle y \rangle_{\text{bottom}}$ to define the engineering shear strain $\gamma = u/H$.

[3 marks]

(c) Plot the relaxed energy and the horizontal reaction force on the top group,

$$R_x = -\sum_{\alpha \in \text{top}} f_x^\alpha, \tag{6}$$

against u (or γ). Here f_x^α is the interatomic force on a top atom *before* it is set to zero in the minimizer. **[3 marks]**

(d) Plot the x position of the core (average x of the core atoms) against u.

(i) At what u does the core first move? From (6) just before that step, estimate the shear stress needed to move the dislocation, $\tau = R_x/W$, where W is the length of the top group.

(ii) What happens to the core, the right surface, the energy and R_x when the core reaches the surface? Show the configuration.

(iii) Why does R_x decrease, and even change sign, while the core moves towards the surface?

[4 marks]

Q4: Neighbour lists and minimization efficiency (8 marks)

Use the triangular lattice and the potential (2) of Q1, with PBC.

(a) Implement the N^2 neighbour search and the Verlet neighbour list with $r_{\text{neigh}} = (1 + \epsilon_{\text{neigh}})r_{\text{cut}}$, building the list with the same N^2 search. Rebuild the list when $\delta_{\max,1} + \delta_{\max,2} > \epsilon_{\text{neigh}} r_{\text{cut}}$, where $\delta_{\max,1}, \delta_{\max,2}$ are the two largest atomic displacements since the last build. Explain why this criterion is safe. **[2 marks]**

(b) Relax `interstitial_10x10.csv`, `interstitial_16x16.csv` and `interstitial_20x20.csv` with CG as in Q1(b), using both searches ($\epsilon_{\text{neigh}} = 0.2$). Compare the total number of pair distances computed (including those computed while building lists), the CPU time, the number of list builds and the final energies. **[3 marks]**

(c) For `interstitial_20x20.csv`, repeat the Verlet run with $\epsilon_{\text{neigh}} = 0.1, 0.2, 0.35$. Report the list length, the number of builds and the pair distances, and explain the trade-off. **[1 mark]**

(d) Relax `vacancy_20x20.csv` with steepest descent and with conjugate gradient to the tolerance of Q1(b). Plot the maximum atomic force against the number of energy/force evaluations and compare iterations, evaluations and final energies. **[2 marks]**

Submission and files

- Submit one report (maximum 8 pages) and the source code with instructions to run it. All plots must come from your code; give figure captions and state units (reduced Lennard–Jones units).
- Python, MATLAB, C or C++ may be used. LAMMPS, other atomistic packages and library minimizers are not allowed; standard numerical and plotting libraries are.
- `parameters.json`: lattice spacings, cell lengths, positions of the removed and inserted atoms, parameters of (5) and the dislocation data (including the ids of the extra half-column).
- `perfect_*.csv`, `vacancy_*.csv`, `interstitial_*.csv` (triangular, 10×10, 16×16, 20×20); `edge_dislocation_initial.csv` (square).
- `README.md`, `check_inputs.py`: file format and basic checks.
- `supplementary_note.pdf`: background material and implementation details.